FINANCIAL TRANSACTIONS

End to End project

FINANCIAL TRANSACTIONS

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1. Data Overview

The Financial Transactions Dataset contains transactional, customer, card, merchant, and fraud-related data used to build a Fraud Detection and Financial Analytics Data Warehouse.

The dataset is designed to support the analysis of customer spending behavior, merchant performance, card usage, and fraudulent transactions through business intelligence and data analytics.

1-Users.csv

Contains customer demographic and financial information, including age, income, credit score, and debt details.
This file is used to build the DimClient table and analyze customer behavior and spending patterns.

Column	Description
id	Unique customer identifier
current_age	Customer's current age
retirement_age	Expected retirement age
birth_year	Customer birth year
birth_month	Customer birth month
gender	Customer gender
address	Customer residential address
latitude	Customer latitude coordinate
longitude	Customer longitude coordinate
per_capita_income	Annual per capita income
yearly_income	Annual customer income
total_debt	Total outstanding debt
credit_score	Customer credit score
num_credit_cards	Number of credit cards owned

2- Cards.csv

Contains detailed information about customer payment cards, such as card type, brand, credit limit, and chip availability.
This file is used to build the **DimCard** table for card portfolio and fraud analysis.

Column	Description
id	Unique card identifier
client_id	Customer identifier
card_brand	Card brand (Visa, Mastercard, etc.)
card_type	Card type (Credit, Debit, Prepaid)
card_number	Card number
expires	Card expiration date
cvv	Card verification value
has_chip	Indicates whether the card contains an EMV chip
num_cards_issued	Number of issued cards
credit_limit	Maximum credit limit
acct_open_date	Card account opening date
year_pin_last_changed	Last PIN update year
card_on_dark_web	Indicates whether the card appeared on the dark web

3- Transactions.csv

Stores all financial transaction records, including transaction amount, merchant, card, customer, and transaction date.
It serves as the primary source for building the FactTransactions table in the data warehouse.

Column	Description
id	Unique transaction identifier
date	Transaction date and time
client_id	Customer identifier
card_id	Card used for the transaction
amount	Transaction amount
use_chip	Indicates whether the chip was used
merchant_id	Merchant identifier
merchant_city	Merchant city
merchant_state	Merchant state
zip	Merchant ZIP code
mcc	Merchant Category Code
errors	Transaction processing errors

4- TrainFraudLabel.json

Contains fraud labels indicating whether each transaction is fraudulent or legitimate.
This file is integrated with transaction data to support fraud detection analysis and reporting.

Column	Description
transaction_id	Transaction identifier
is_fraud	Fraud label (Yes / No)

5- MCC.json

Provides the mapping between Merchant Category Codes (MCC) and their business descriptions.
It is used to create the DimMCC table for analyzing transaction and fraud trends by merchant category.

Column	Description
mcc_code	Merchant Category Code
mcc_description	Merchant category description

1.1 Analytical Questions

Q1: What was the total transaction amount and transaction count by month over the last 12 months?
Dimensions Used: DimDate. Business Value:
متابعة نمو حجم المعاملات واكتشاف الاتجاهات الموسمية.

Q2: Which merchant categories (MCC) generated the highest transaction value during the last quarter?
Dimensions Used: DimDate + DimMCC.
Business Value: معرفة أكثر الأنشطة التجارية تحقيقًا للإيرادات وتوجيه الشراكات التجارية.

Q3: Which cities and states recorded the highest fraud rate during the past 6 months?
Dimensions Used: DimDate + DimMerchant. Business Value:
تحديد المناطق الجغرافية الأكثر خطورة وتحسين سياسات إدارة المخاطر.

Q4: How did transaction amounts vary across different customer income groups during the current year?
Dimensions Used: DimDate + DimClient.
Business Value: فهم سلوك الإنفاق حسب مستوى الدخل وتطوير عروض مناسبة لكل شريحة.

Q5: Which card brands and card types experienced the highest fraud rate in the previous year?
Dimensions Used: DimDate + DimCard. Business Value: تقييم مخاطر كل نوع بطاقة واتخاذ قرارات أمنية وتشغيلية.

Q6: What are the monthly fraud trends across merchant categories over the last 12 months?
Dimensions Used: DimDate + DimMCC.
Business Value:
اكتشاف تغيرات الاحتيال حسب النشاط التجاري مع مرور الوقت.

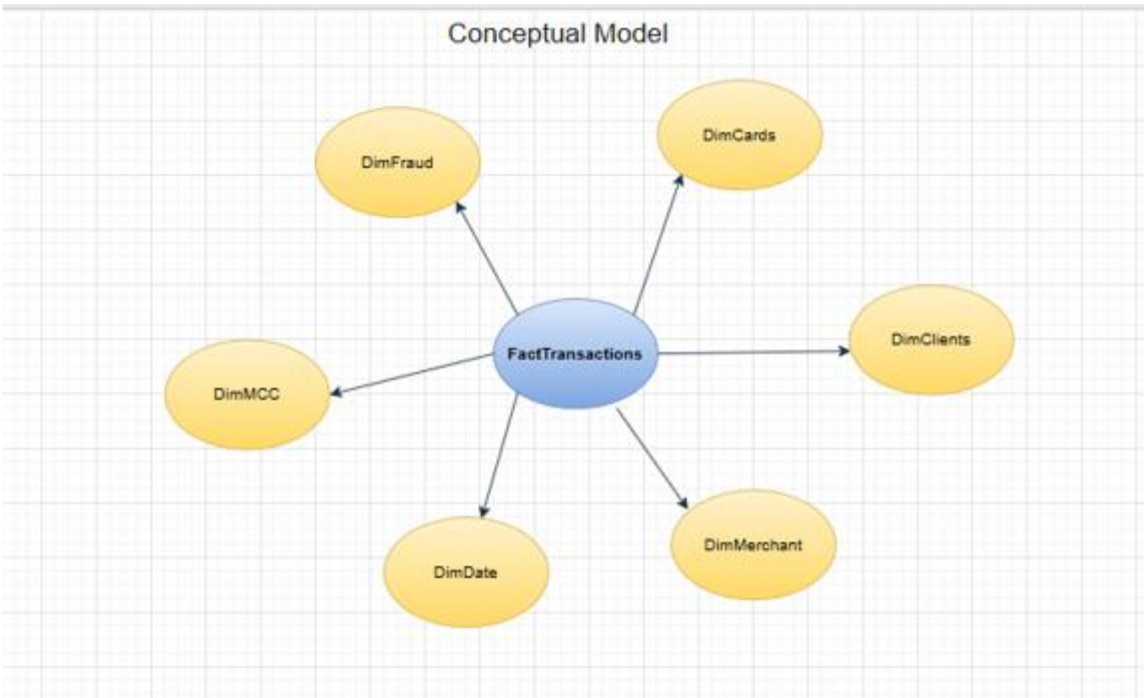
Q7: Which customer age groups generated the highest transaction volume in the last quarter?
Dimensions Used: DimDate + DimClient.
BusinessValue:
فهم الشرائح العمرية الأكثر نشاطًا لتوجيه الحملات التسويقية و الخدمات.

Q8: How has the average transaction amount changed by merchant state during the past year?
Dimensions Used: DimDate + DimMerchant.
Business Value: مقارنة أداء الولايات المختلفة وتحديد الأسواق الأكثر نشاطًا.

Q9: What percentage of fraudulent transactions was performed using chip-enabled versus non-chip cards during the last 6 months?
Dimensions Used: DimDate + DimCard.
Business Value:
قياس فعالية تقنية الـ chip
في الحد من الاحتيال.

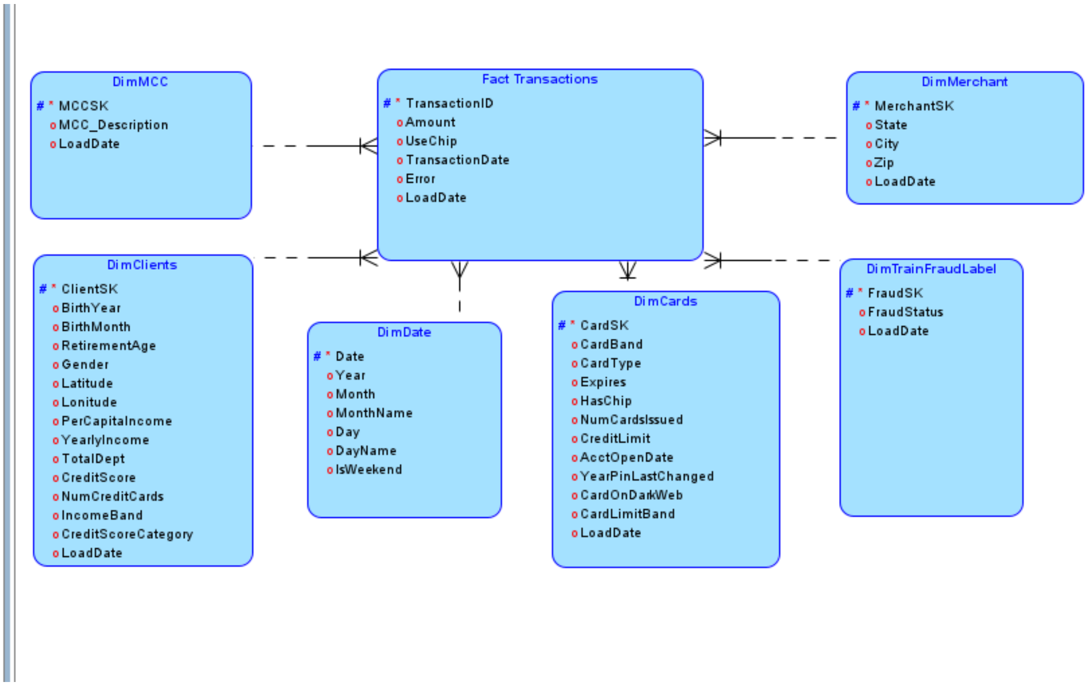
2. Conceptual, Logical & Physical Model

The conceptual model illustrates the overall business structure of the data warehouse. It identifies the main fact table (FactTransactions) and its related dimension tables, providing a clear view of how transactional data is organized for analytical reporting and fraud analysis.



2.1 Logical Model

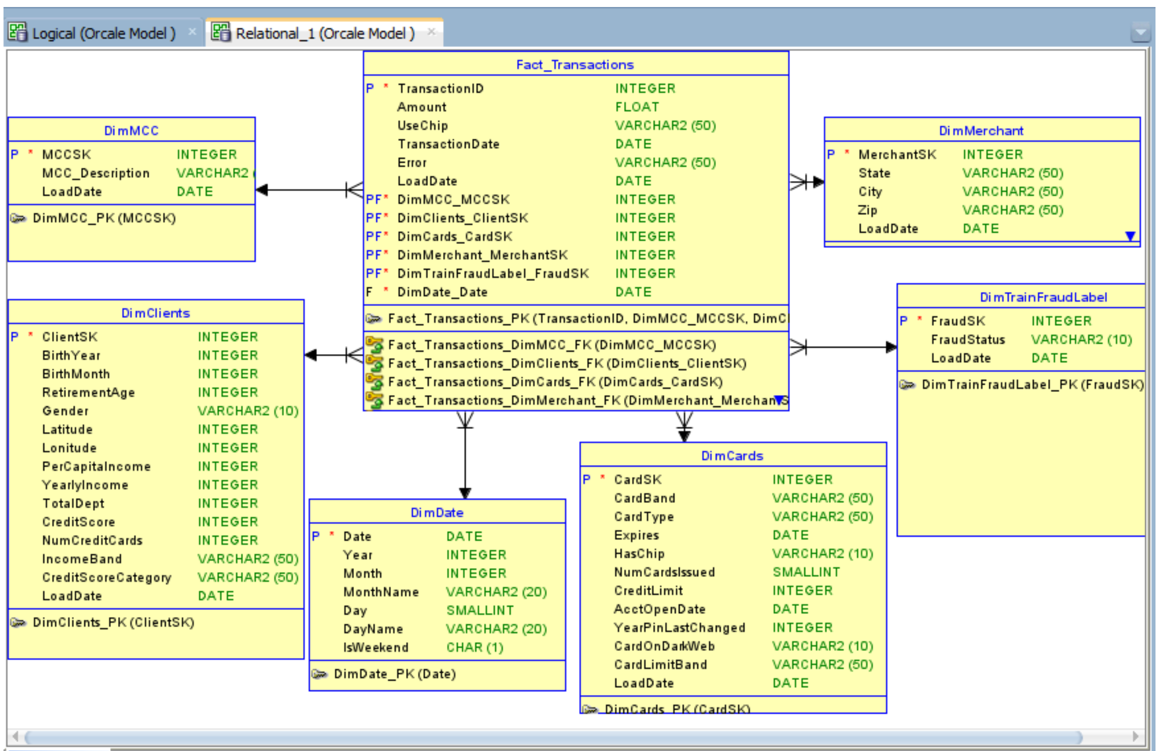
The logical model represents the detailed structure of the data warehouse by defining entities, attributes, keys, and relationships. It serves as the foundation for the physical implementation, ensuring data consistency and supporting efficient business intelligence and fraud analytics.



2.2 Physical Model

Implements the logical design as a SQL Server database by defining tables, columns, data types, primary keys, foreign keys, and constraints to support ETL, SSAS Tabular, and Power BI analytics.

The physical model describes the final database implementation by defining tables, columns, data types, primary and foreign keys, and database constraints. It provides the optimized structure required for loading, storing, and analyzing transactional data within the data warehouse environment.



Description of Tables:

• FactTransactions (Fact Table):
Stores detailed financial transaction records, including transaction amount, transaction date, chip usage, processing errors, and references to customers, cards, merchants, dates, and merchant categories.

• DimClients (Customer Dimension):
Contains customer demographic and financial information such as age, gender, income, debt, and credit score for customer segmentation and behavioral analysis.

• DimCards (Card Dimension):
Stores card information including card brand, type, credit limit, chip availability, and other attributes used for card performance and fraud analysis.

• DimMerchant (Merchant Dimension):
Contains merchant location details including city, state, and ZIP code, enabling geographical analysis of transactions.

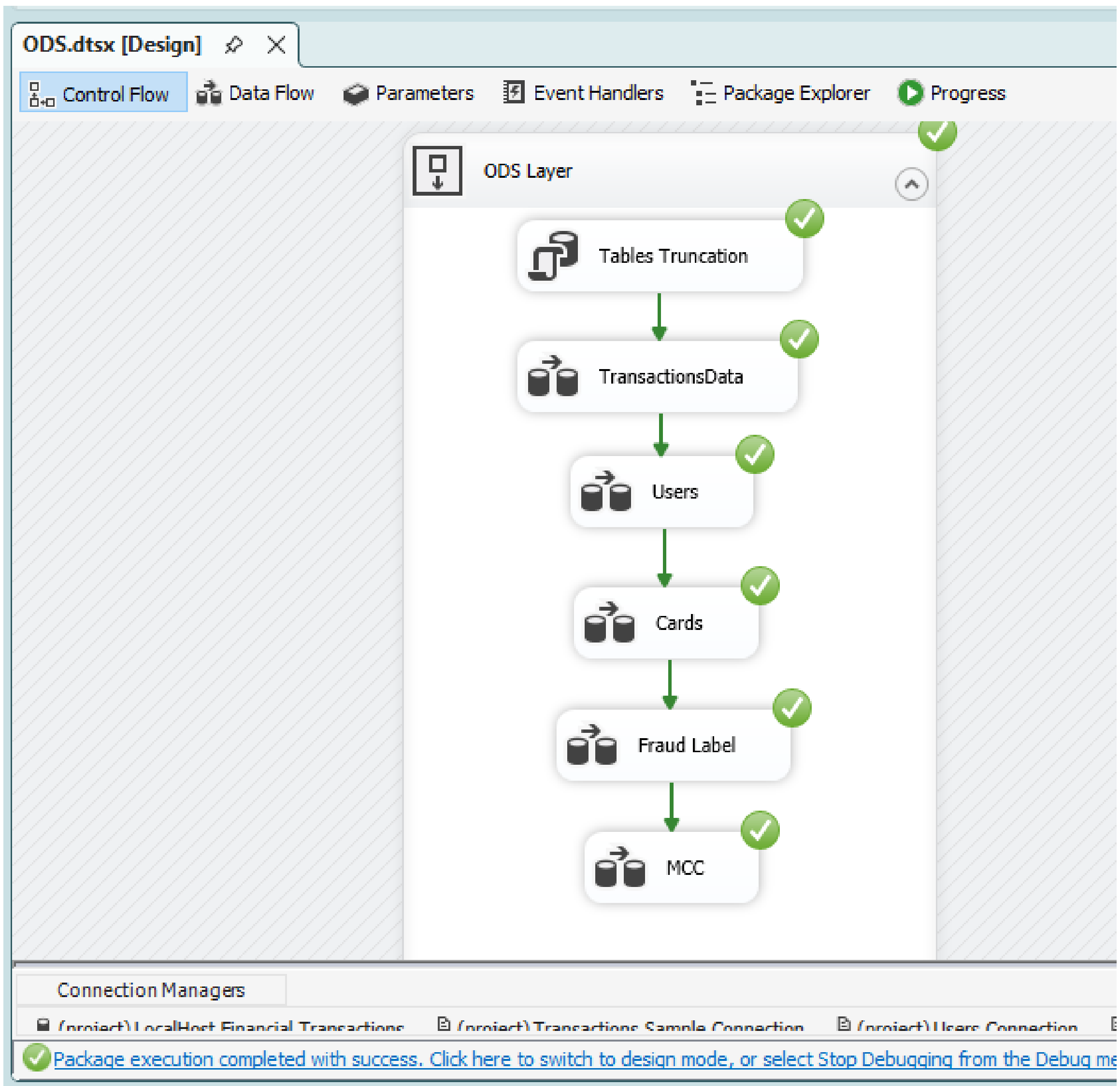
• DimMCC (Merchant Category Dimension):
Provides Merchant Category Code (MCC) descriptions used to classify merchants and analyze transactions by business category.
• DimDate (Date Dimension):
Contains calendar attributes such as year, month, day, and weekend indicator to support time-based analysis and reporting.
• DimFraud (Fraud Dimension):
Contains fraud status labels used to identify fraudulent transactions and calculate fraud-related metrics.

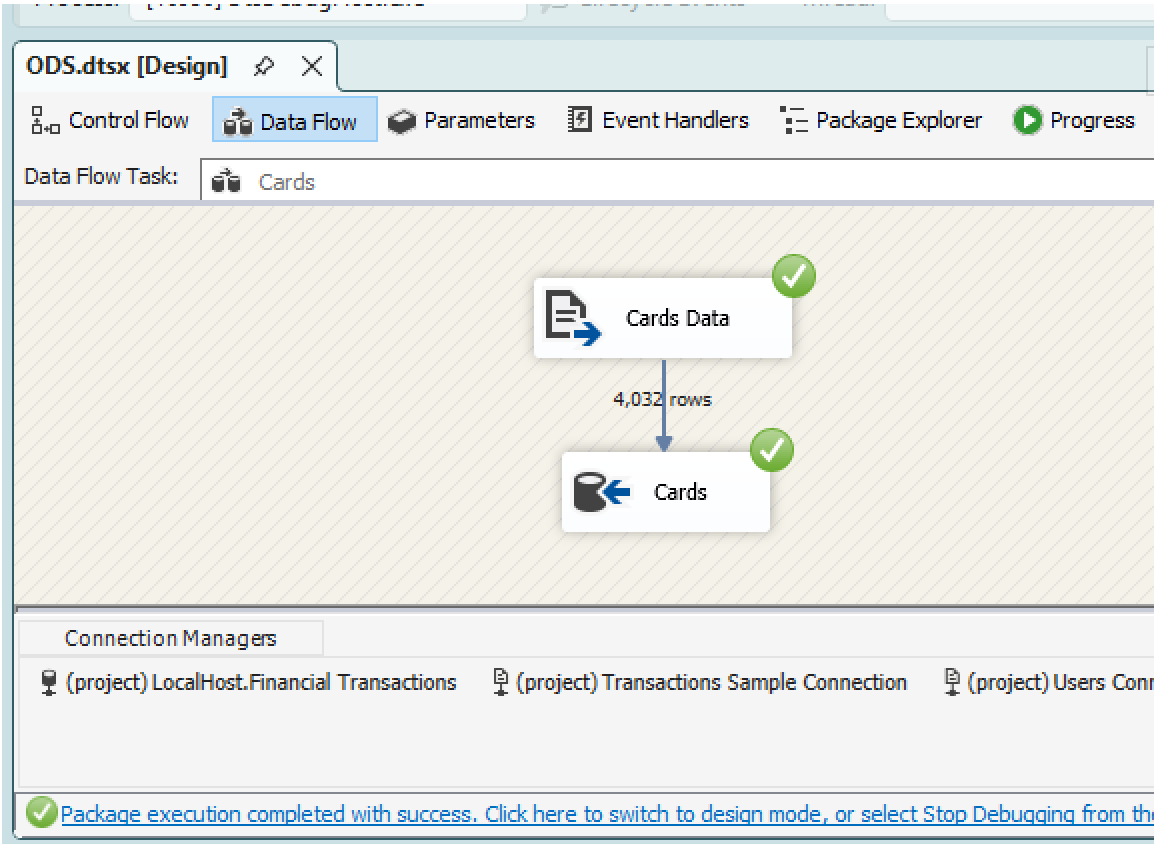
3. SSIS Overview

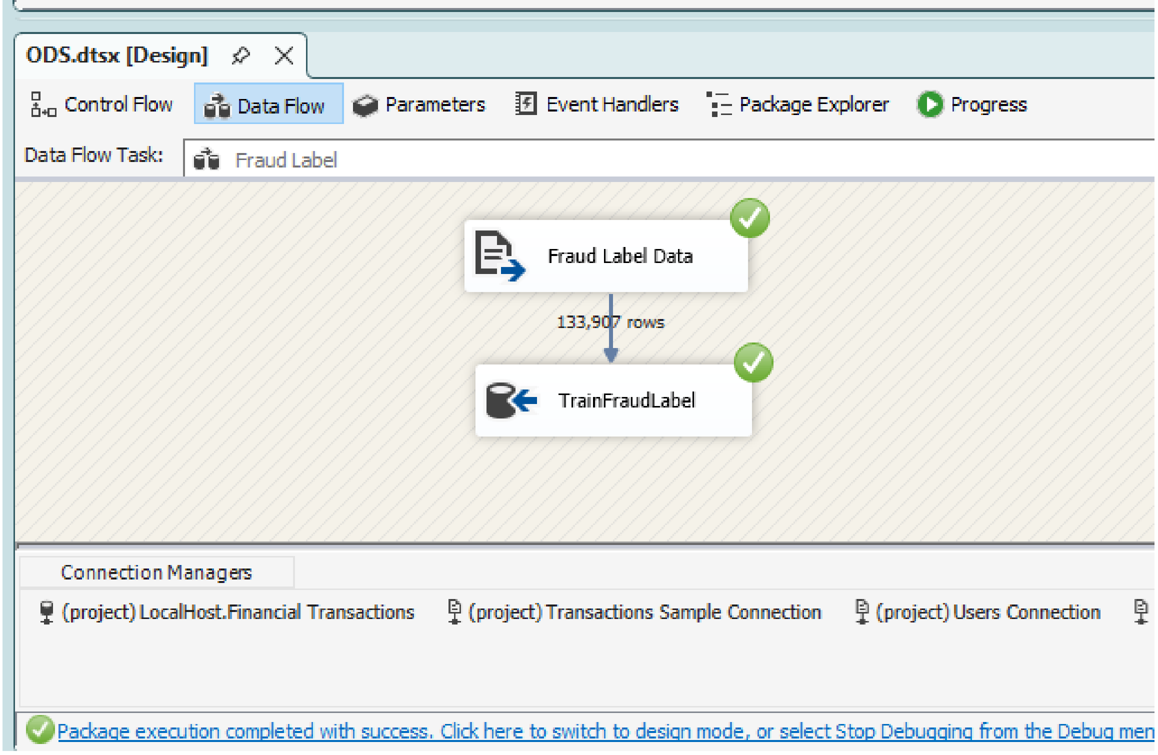
The ETL process was implemented using SQL Server Integration Services (SSIS) to extract data from multiple source files, perform data cleansing and business transformations, and load the data into a Star Schema Data Warehouse.
The ETL process consists of three layers:

- Operational Data Store (ODS)
- Staging Area (STG)
- Data Warehouse (DWH)

1- **ODS Layer**
The raw CSV and JSON files were imported into SQL Server without modifications.




ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Cards
Cards Data
4,032 rows
Cards
Connection Managers
(project) LocalHost.Financial Transactions
(project) Transactions Sample Connection
(project) Users Conn
Package execution completed with success. Click here to switch to design mode, or select Stop Debugging from th


ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Fraud Label
Fraud Label Data
133,907 rows
TrainFraudLabel
Connection Managers
(project) LocalHost.Financial Transactions
(project) Transactions Sample Connection
(project) Users Connection
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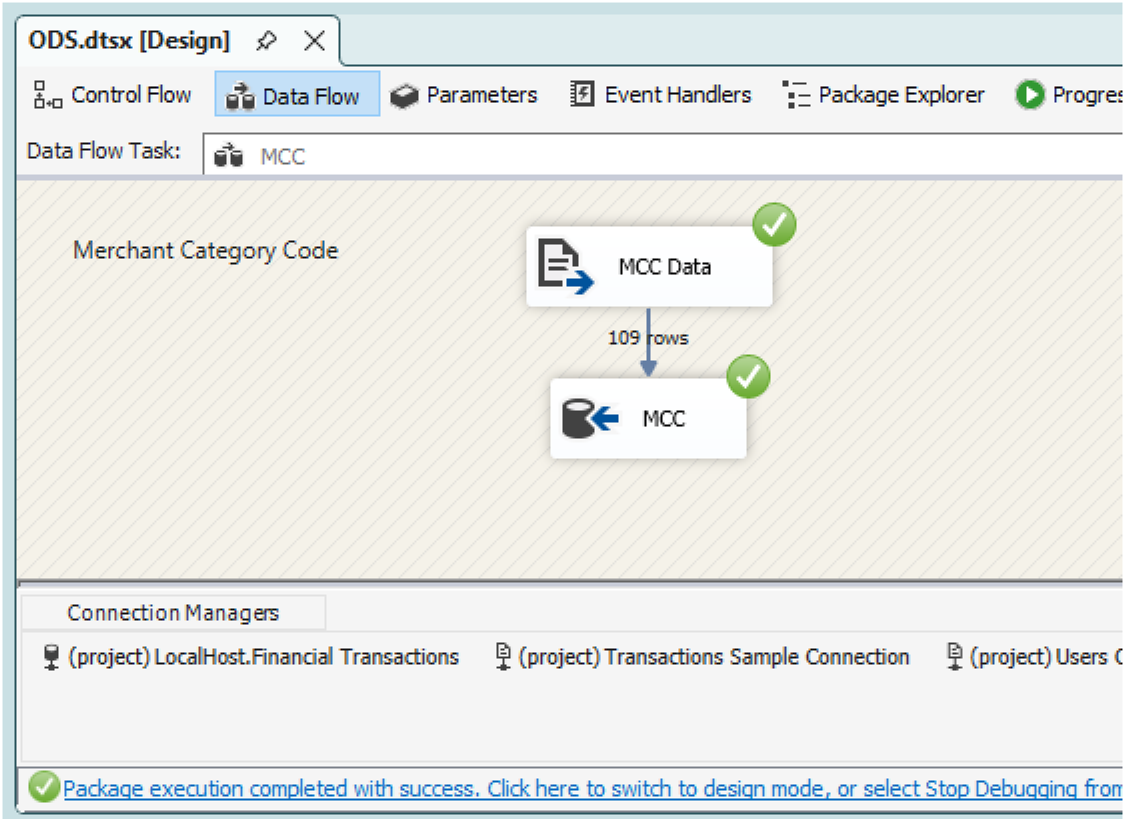
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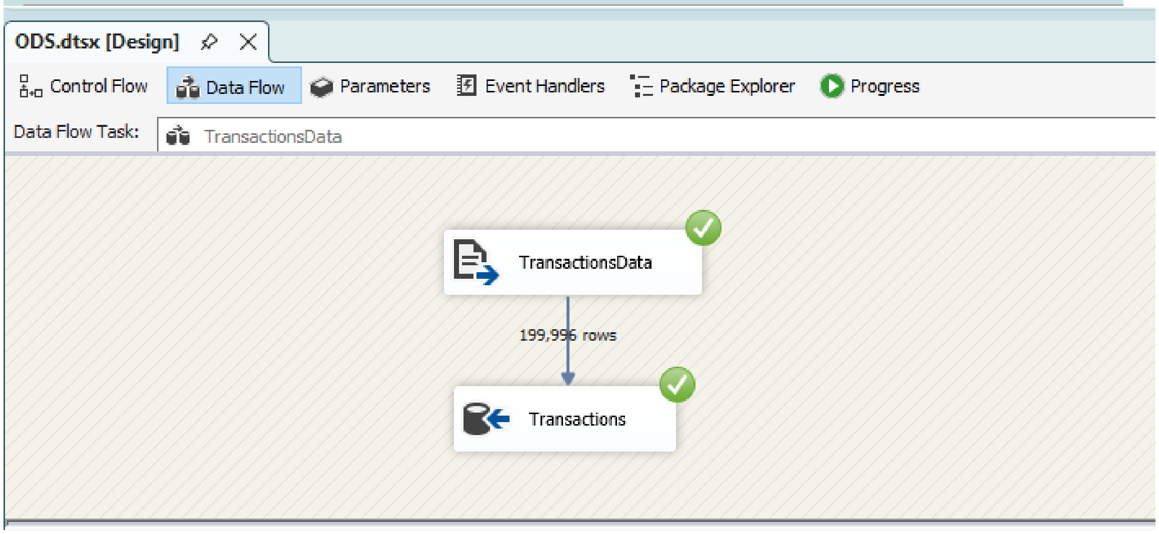
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Data Flow Task:
MCC
Merchant Category Code
MCC Data
109 rows
MCC
Connection Managers
(project) LocalHost.Financial Transactions
(project) Transactions Sample Connection
Package execution completed with success. Click here to switch to design mode, or select Stop Debugging from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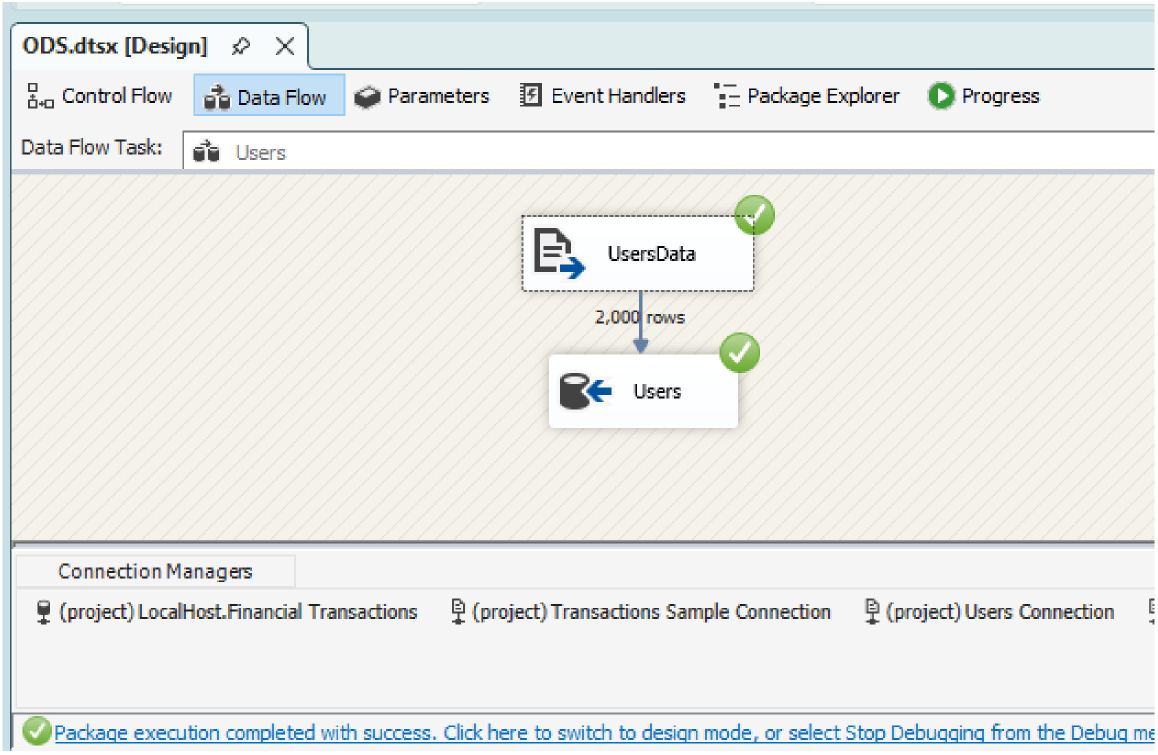
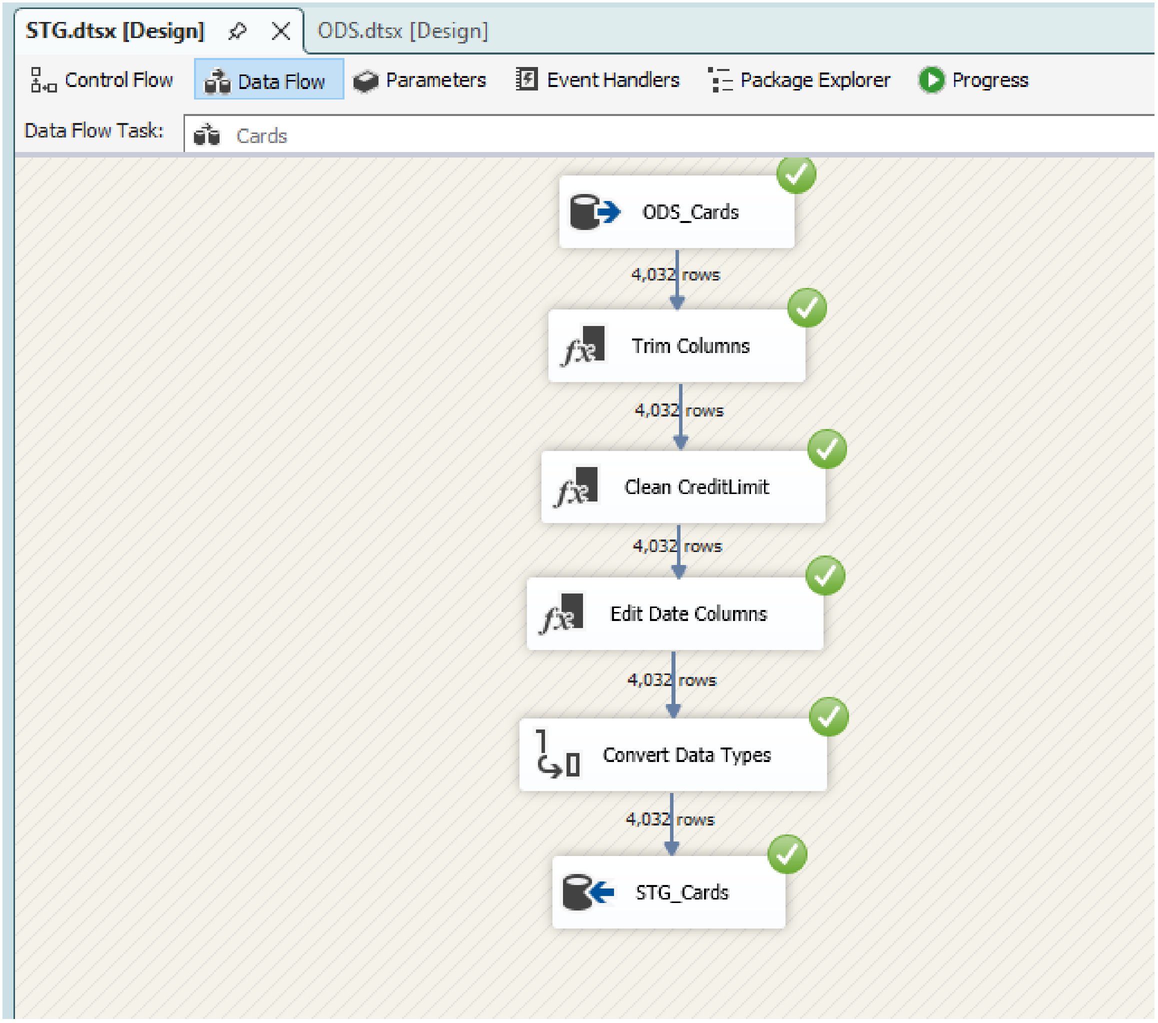
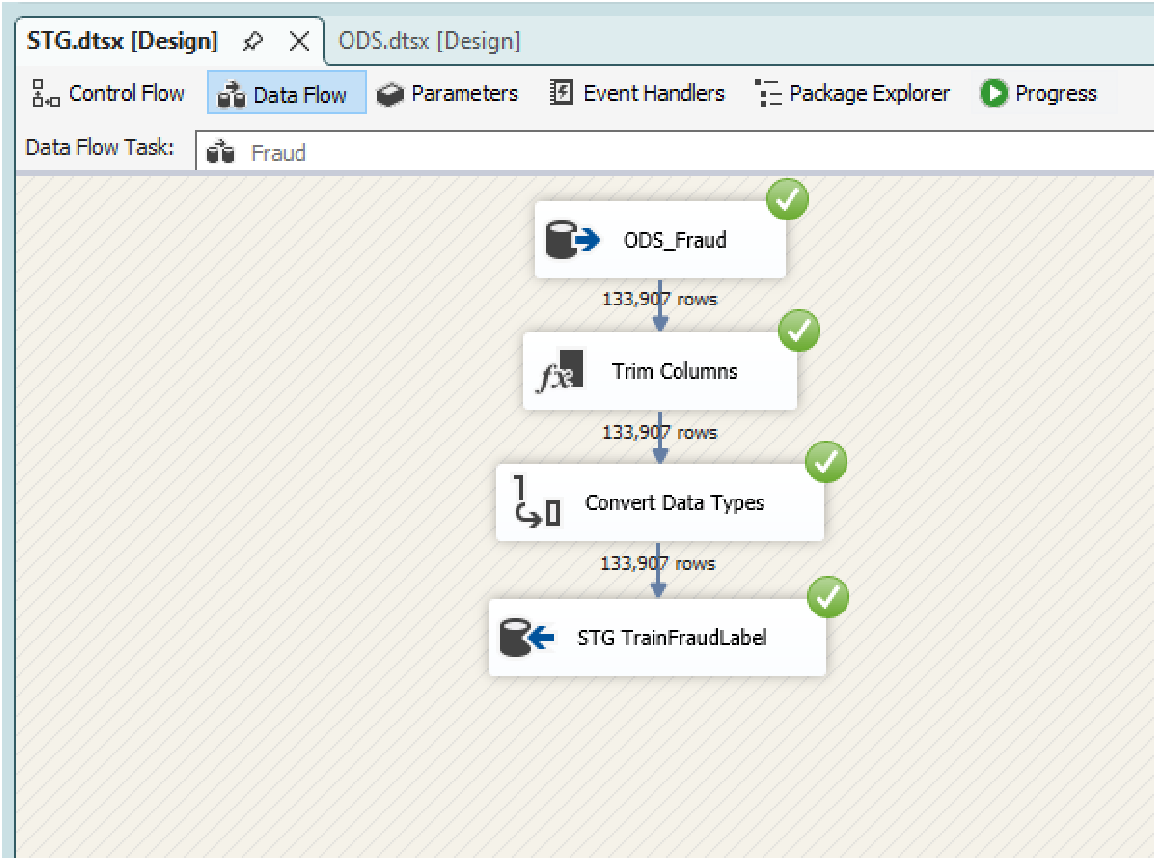
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
TransactionsData
TransactionsData
199,996 rows
Transactions

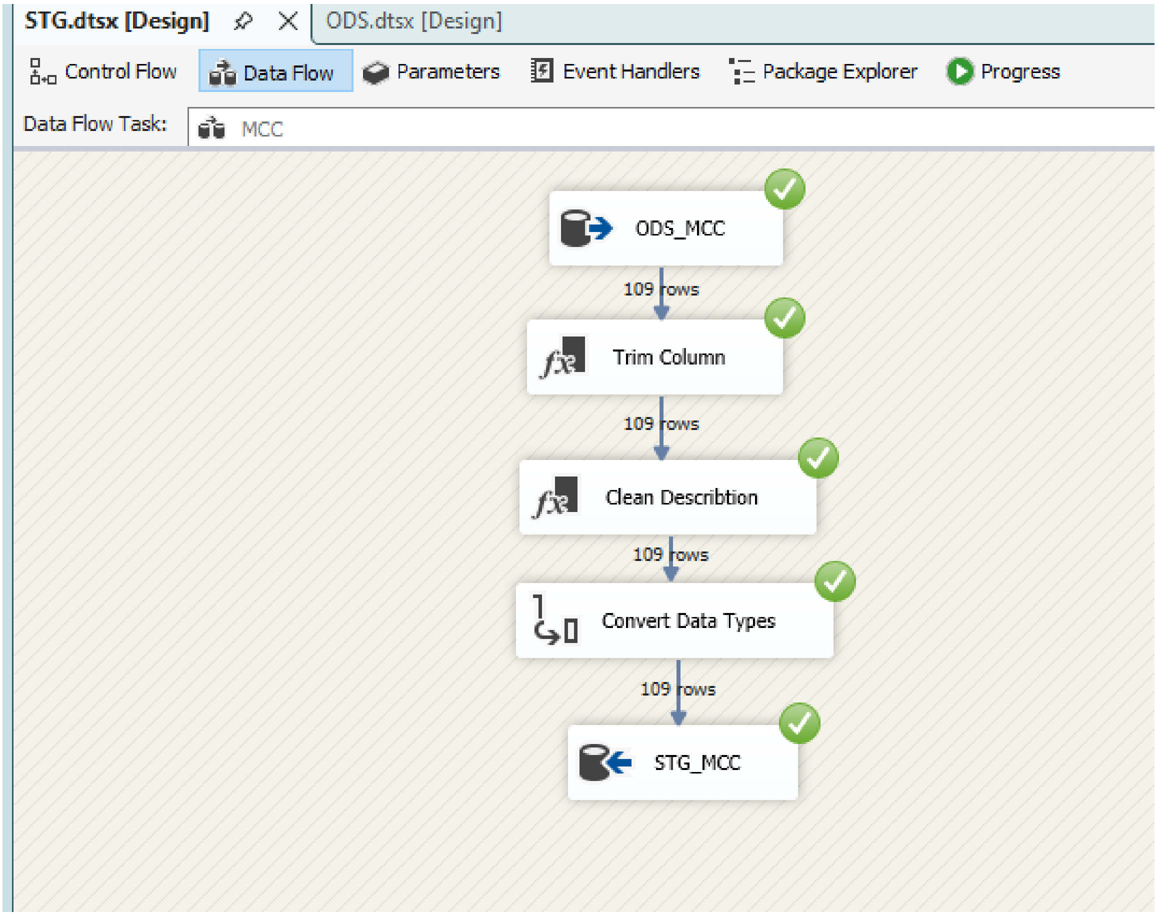


2- STG

Data cleansing and transformations were applied including.


STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Cards
ODS_Cards
4,032 rows
Trim Columns
4,032 rows
Clean CreditLimit
4,032 rows
Edit Date Columns
4,032 rows
Convert Data Types
4,032 rows
STG_Cards


STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Fraud
ODS_Fraud
133,907 rows
Trim Columns
133,907 rows
Convert Data Types
133,907 rows
STG TrainFraudLabel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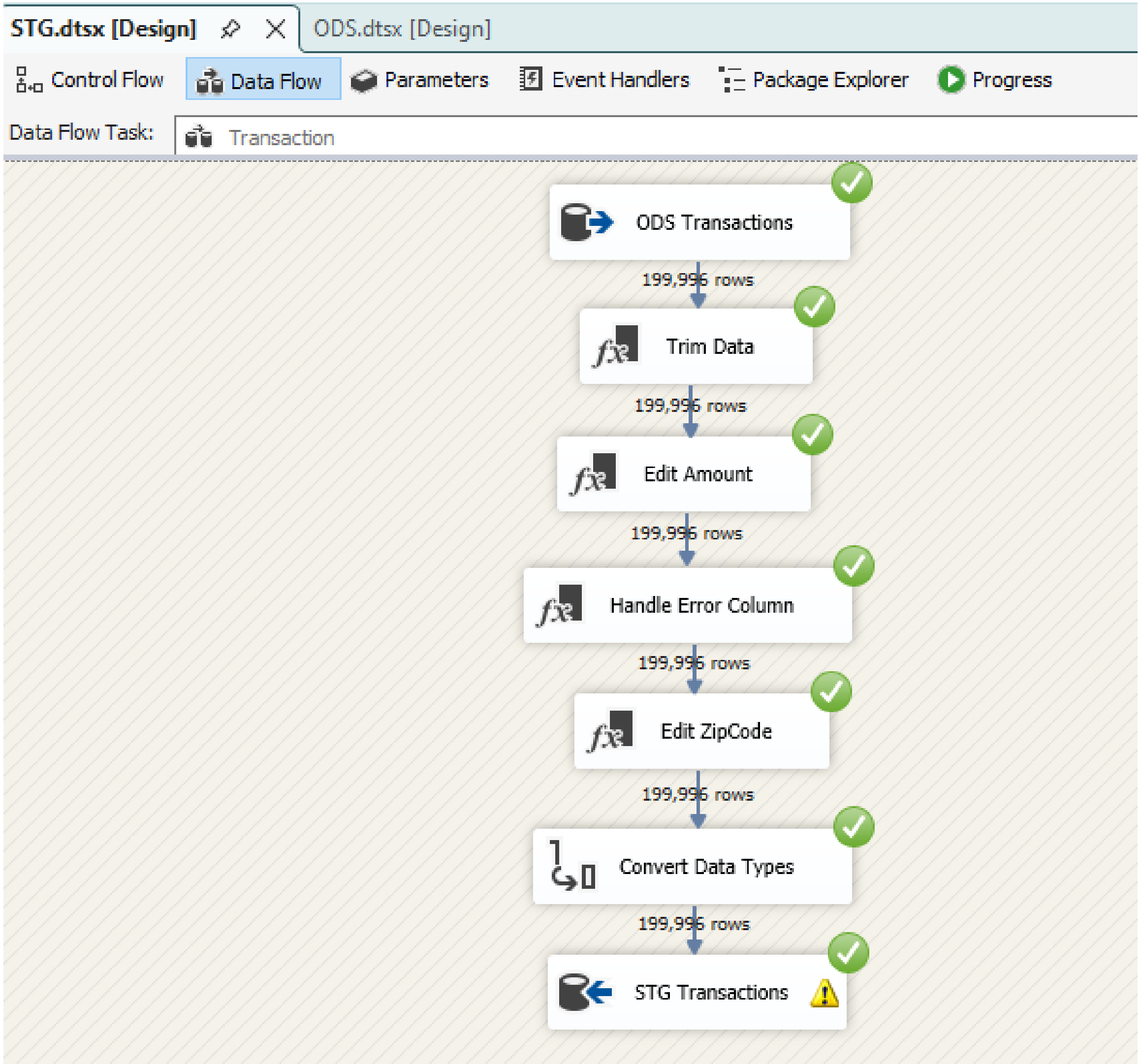
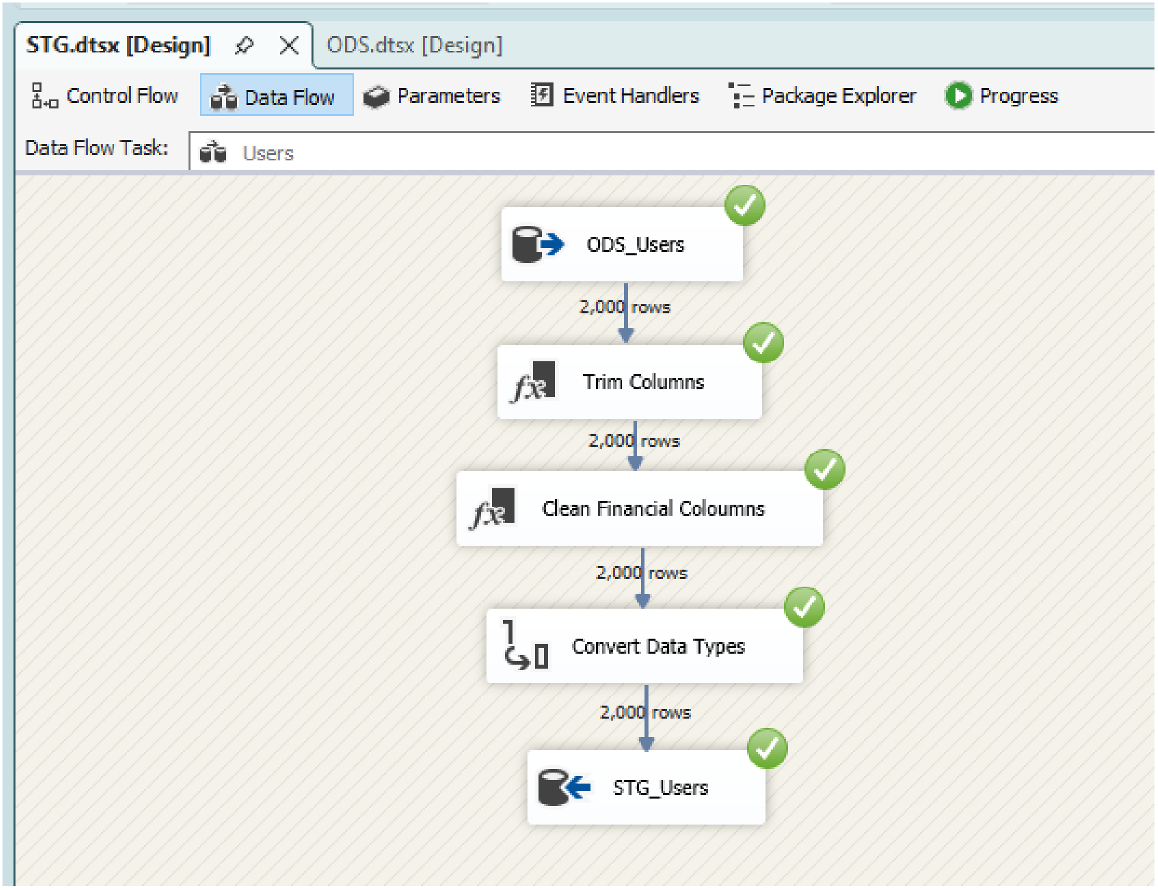
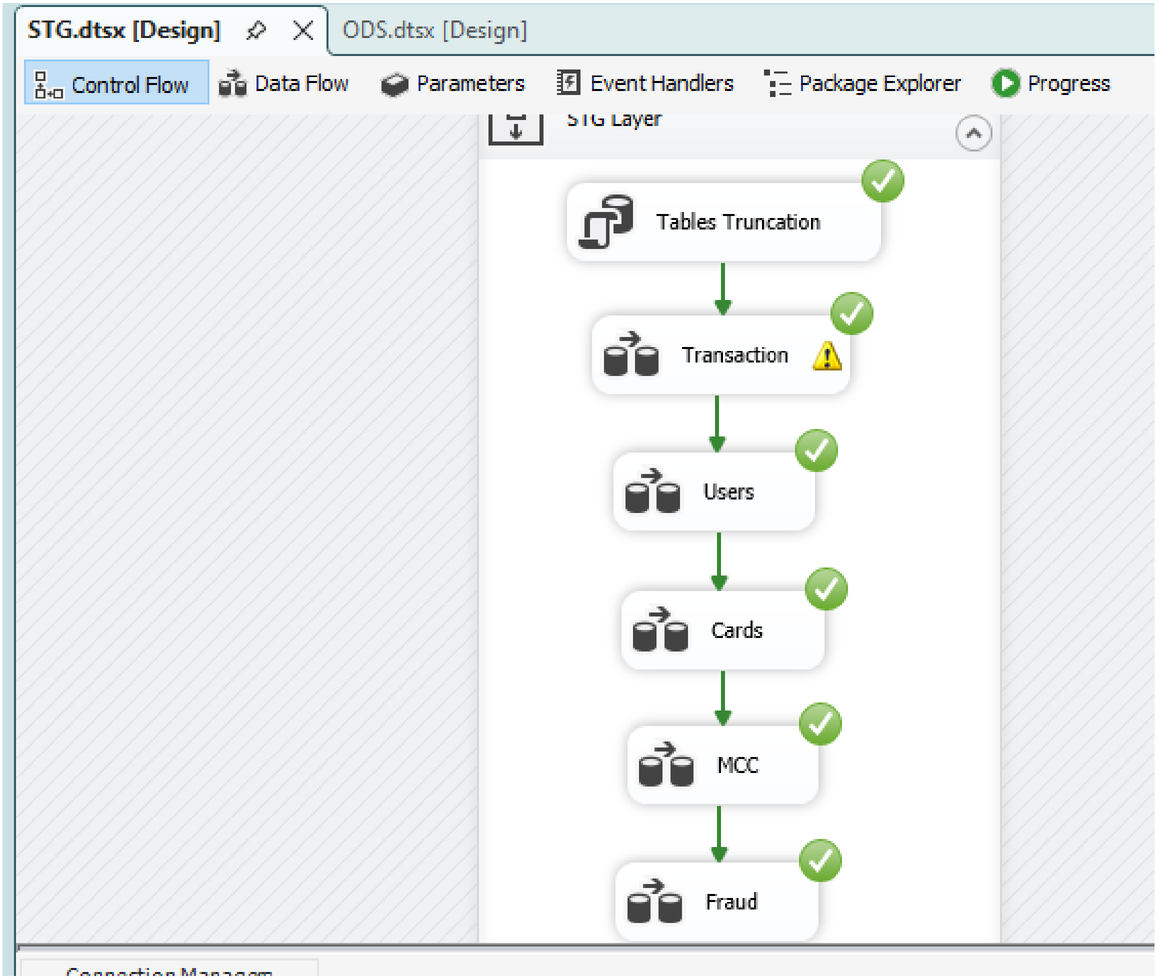
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
MCC
ODS_MCC
109 rows
Trim Column
109 rows
Clean Describtion
109 rows
Convert Data Types
109 rows
STG_MCC

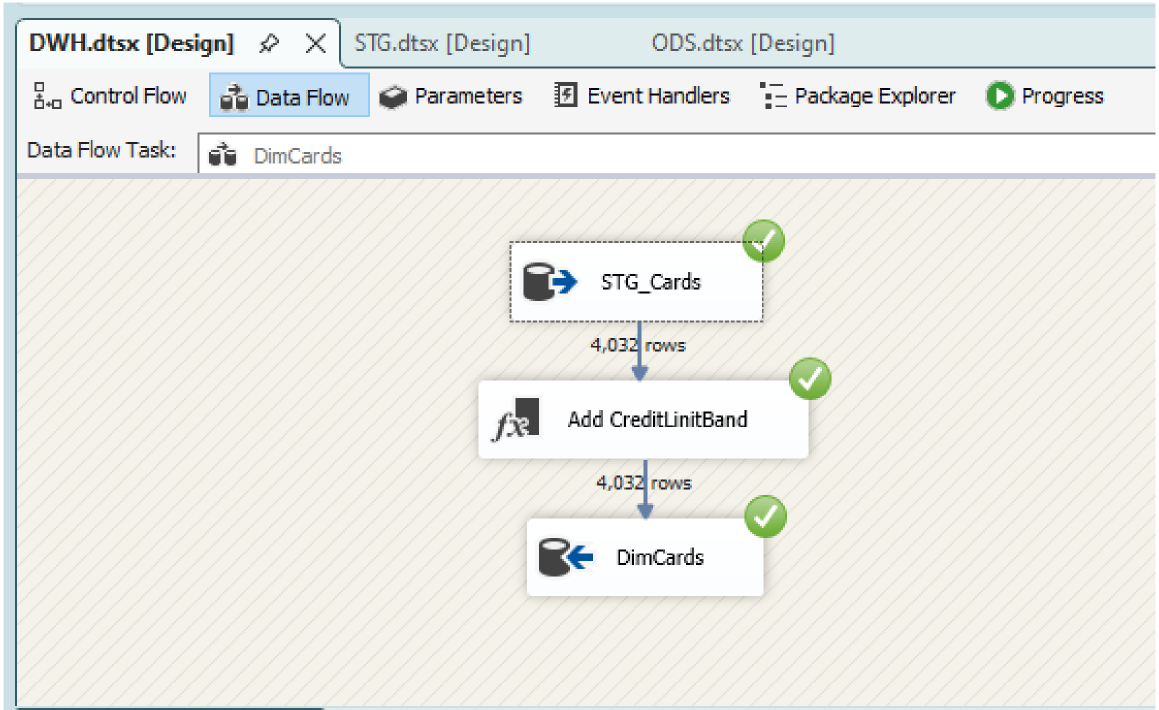

STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Transaction
ODS Transactions
199,996 rows
Trim Data
199,996 rows
Edit Amount
199,996 rows
Handle Error Column
199,996 rows
Edit ZipCode
199,996 rows
Convert Data Types
199,996 rows
STG Transactions

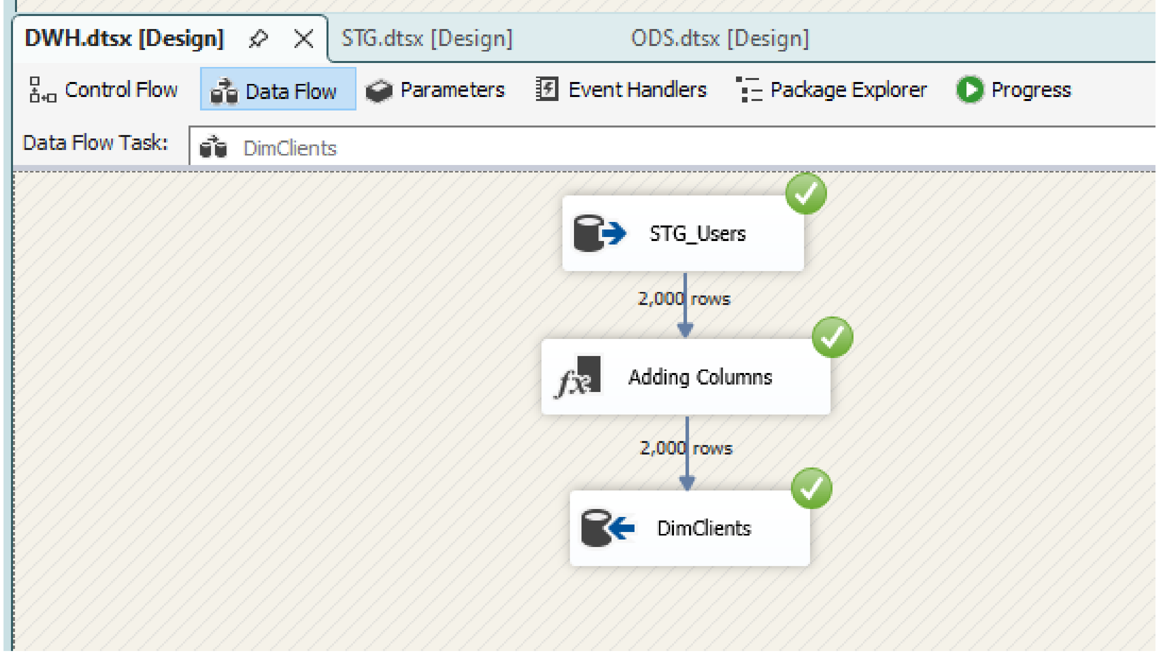

STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Users
ODS_Users
2,000 rows
Trim Columns
2,000 rows
Clean Financial Coloumns
2,000 rows
Convert Data Types
2,000 rows
STG_Users

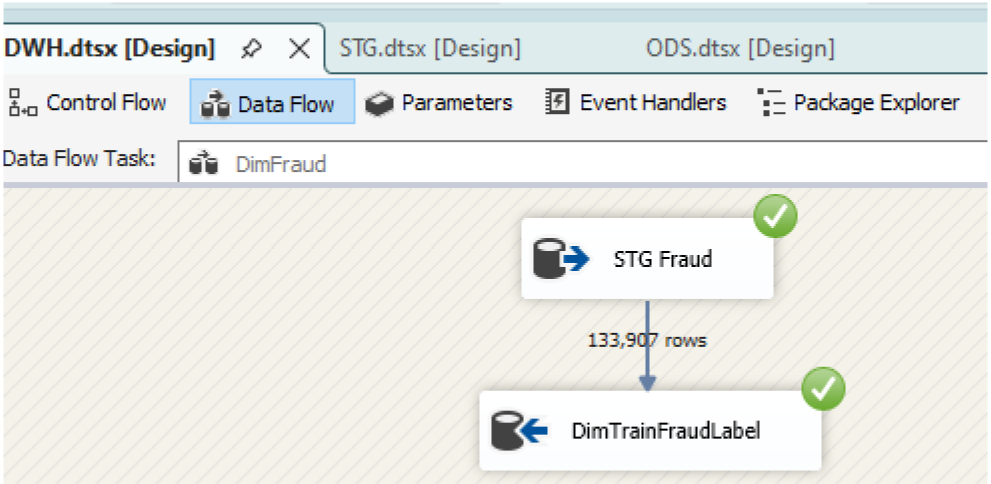


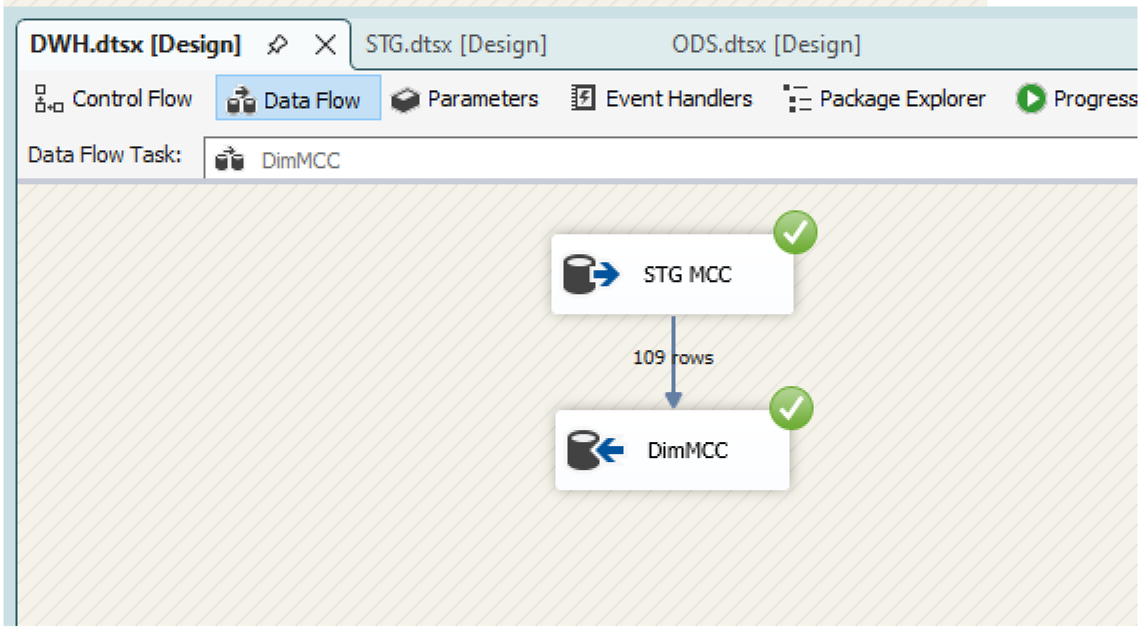
3- DWH

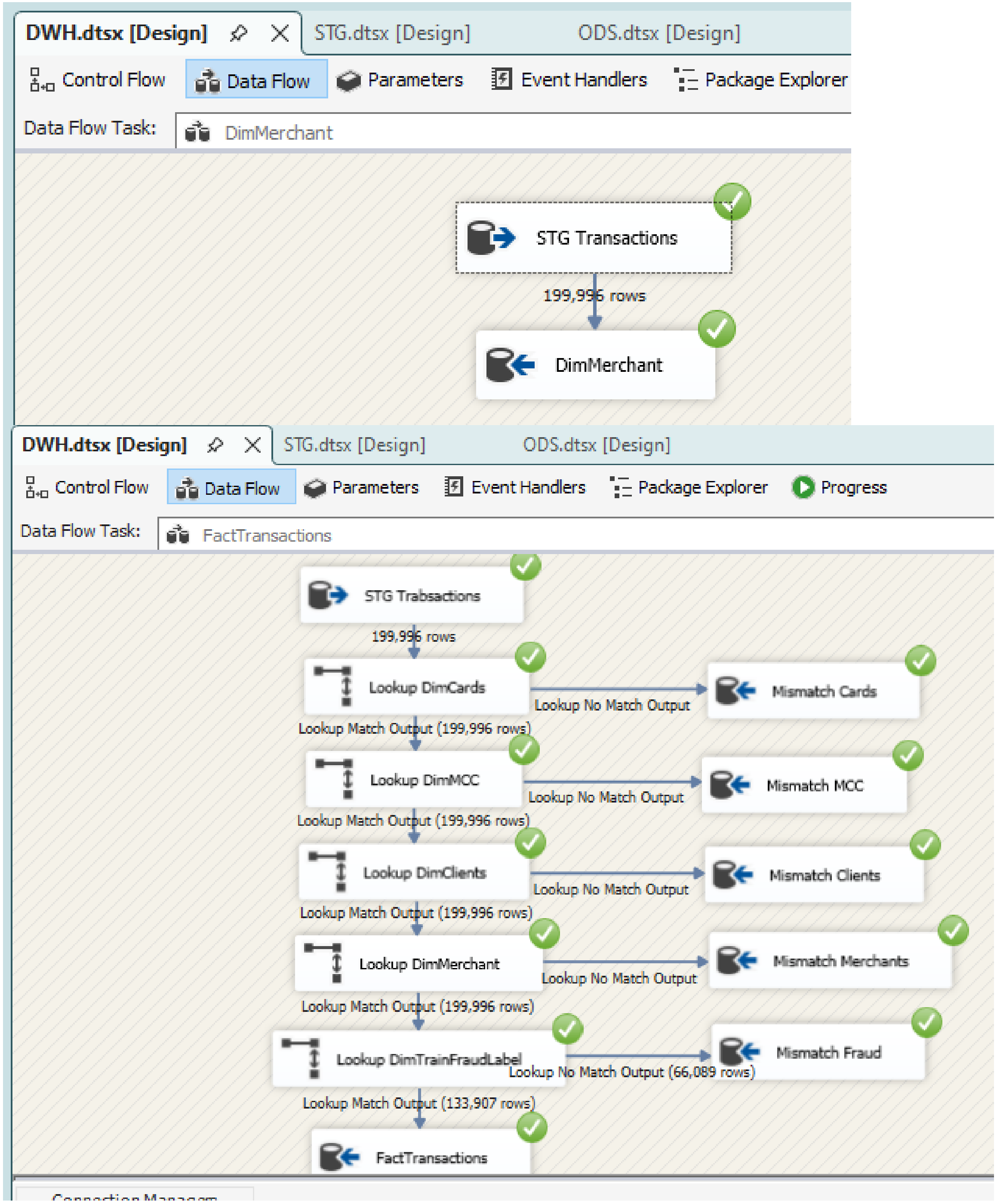
Dimension tables were loaded first followed by the FactTransactions table using Lookup transformations to retrieve surrogate keys


DWH.dtsx [Design]
STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
DimCards
STG_Cards
4,032 rows
Add CreditLinitBand
4,032 rows
DimCards


DWH.dtsx [Design]
STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
DimClients
STG_Users
2,000 rows
Adding Columns
2,000 rows
DimClients


DWH.dtsx [Design]
STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Data Flow Task:
DimFraud
STG Fraud
133,907 rows
DimTrainFraudLabel

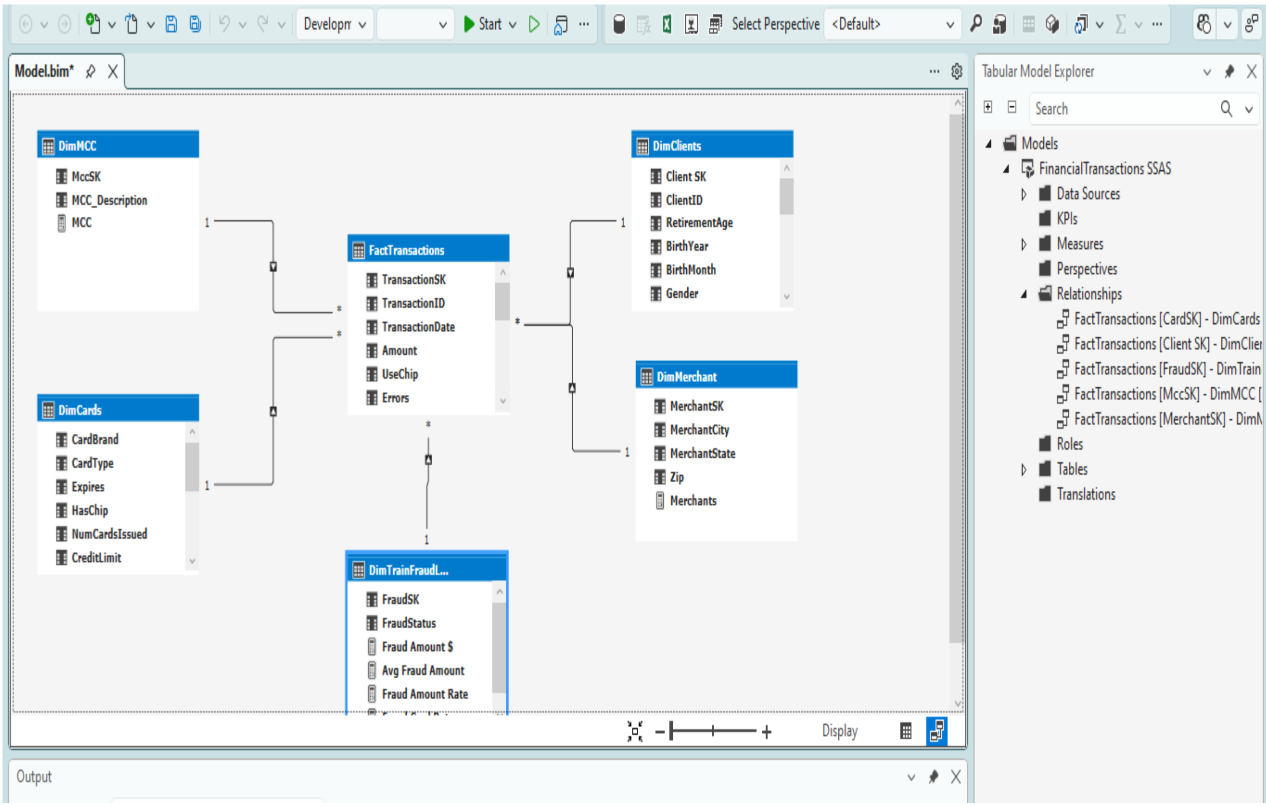

DWH.dtsx [Design]
STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
DimMCC
STG MCC
109 rows
DimMCC



DWH.dtsx [Design]
STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Data Flow Task:
DimMerchant
STG Transactions
199,996 rows
DimMerchant
DWH.dtsx [Design]
STG.dtsx [Design]
ODS.dtsx [Design]
Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
FactTransactions
STG Trabsactions
199,996 rows
Lookup DimCards
Lookup No Match Output
Mismatch Cards
Lookup Match Output (199,996 rows)
Lookup DimMCC
Lookup No Match Output
Mismatch MCC
Lookup Match Output (199,996 rows)
Lookup DimClients
Lookup No Match Output
Mismatch Clients
Lookup Match Output (199,996 rows)
Lookup DimMerchant
Lookup No Match Output
Mismatch Merchants
Lookup Match Output (199,996 rows)
Lookup DimTrainFraudLabel
Lookup No Match Output (66,089 rows)
Mismatch Fraud
Lookup Match Output (133,907 rows)
FactTransactions

4. SSAS Overview

The Data Warehouse was imported into SQL Server Analysis Services (SSAS) Tabular to build a semantic model optimized for reporting and interactive analysis. Relationships were created between the fact and dimension tables, followed by the implementation of calculated columns and DAX measures.



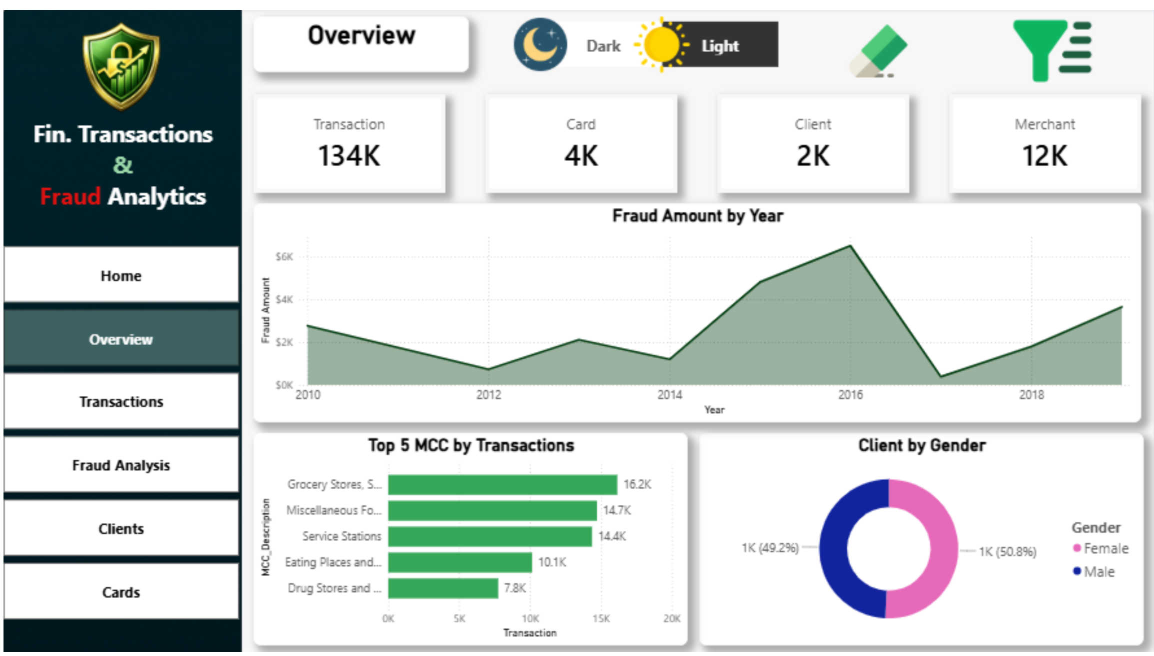
5. Power BI Overview

The Power BI report connects directly to the SSAS Tabular model to provide interactive dashboards for transaction analysis, fraud monitoring, customer behavior, and card performance.

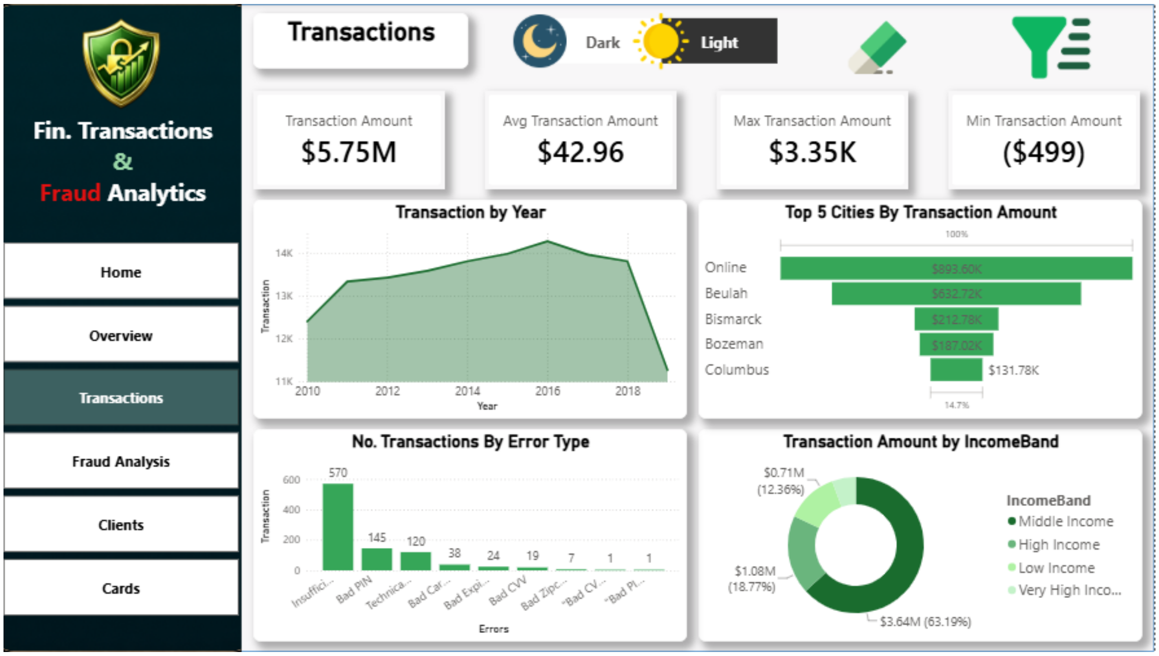
Measure Name	Dax Expression
Avg Transaction Amount	AVERAGE(FactTransactions[Amount])
Max Transaction Amount	MAX(FactTransactions[Amount])
Min Transaction Amount	MIN(FactTransactions[Amount])
Transaction Amount	SUM(FactTransactions[Amount])
# Cards	COUNT(DimCards[CardSK])
# Clients	COUNT(DimClients[Client SK])
# Merchants	COUNT(FactTransactions[MerchantSK])
# Transactions	COUNTROWS(FactTransactions)
Avg Fraud Amount	CALCULATE(AVERAGE(FactTransactions[Amount]), FactTransactions [FraudStatus] = "YES")
Fraud Amount	CALCULATE([Transaction Amount], FactTransactions[FraudStatus] = "YES")
Fraud Amount Rate	DIVIDE([Fraud Amount],[Transaction Amount])
Fraud Card Rate	DIVIDE([Fraud Cards],[Card])
Fraud Cards	CALCULATE(DISTINCTCOUNT(FactTransactions[CardSK]),DimTrainFraudLabel[FraudStatus] = "YES")
Fraud Clients	CALCULATE(DISTINCTCOUNT(FactTransactions[Client SK]),DimTrainFraudLabel[FraudStatus] = "YES")
Fraud Clients Rate	DIVIDE([Fraud Clients],[Client])
Fraud Merchants	CALCULATE(DISTINCTCOUNT(FactTransactions[MerchantSK]), DimTrainFraudLabel[FraudStatus] = "YES")
Fraud Rate	DIVIDE([Fraud Transactions],[Transaction])
Fraud Transactions	CALCULATE([Transaction],DimTrainFraudLabel[FraudStatus] = "YES")
Merchant Fraud Rate	DIVIDE([Fraud Merchants],[Merchant])

1- Home Landing Page.

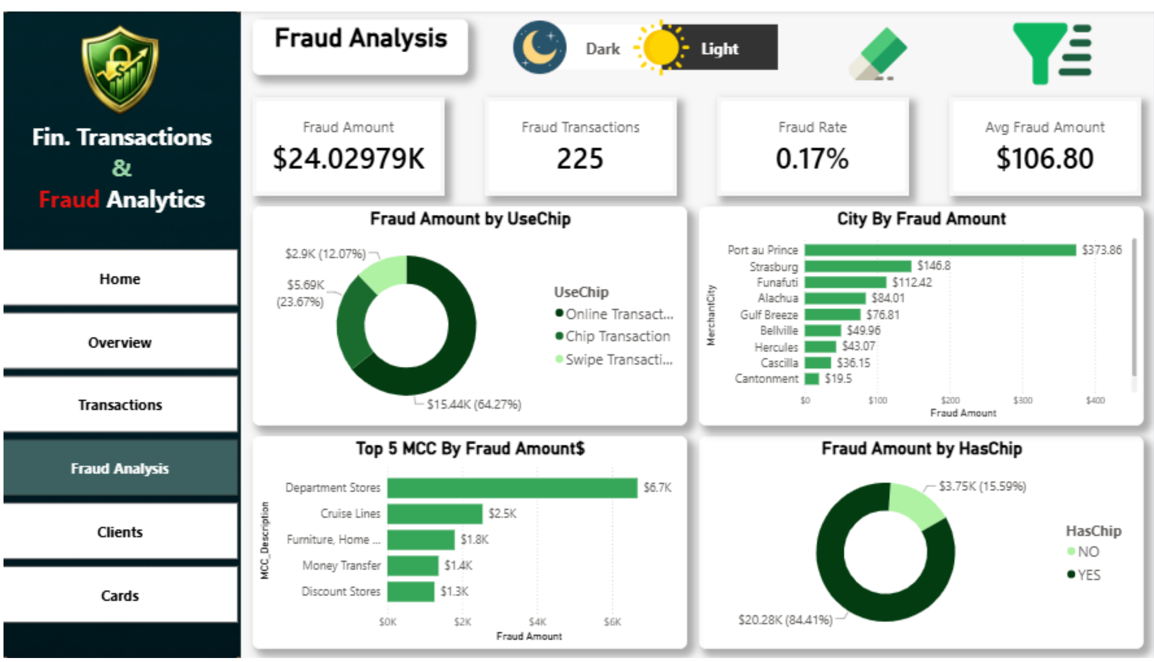
2- Overview



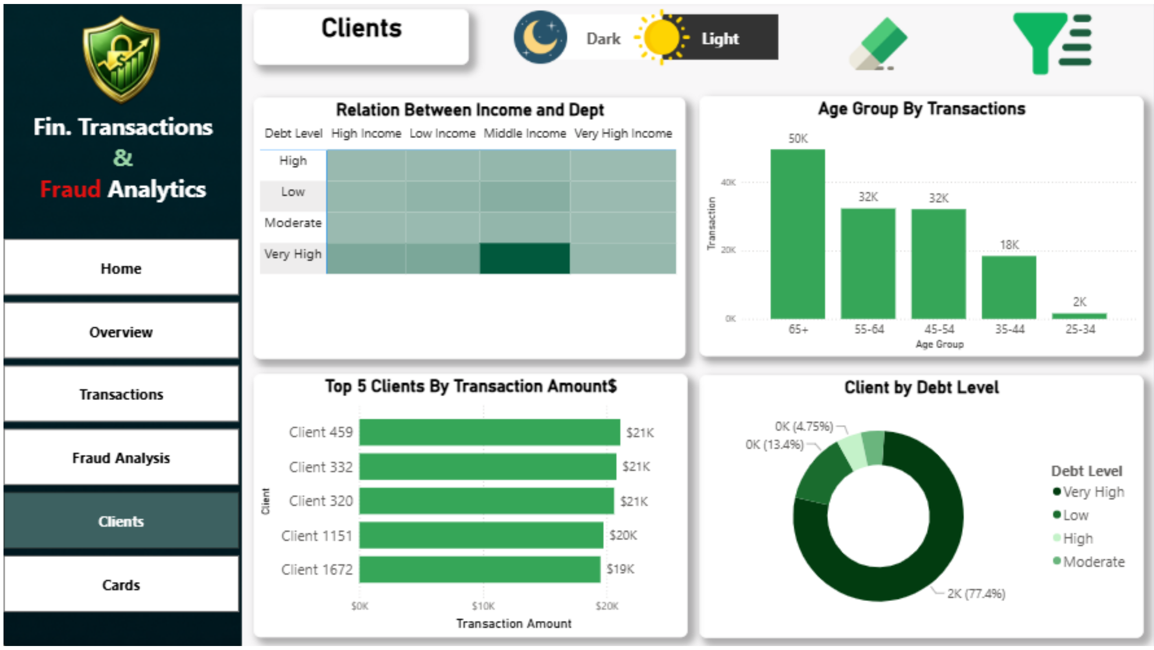
3- Transactions



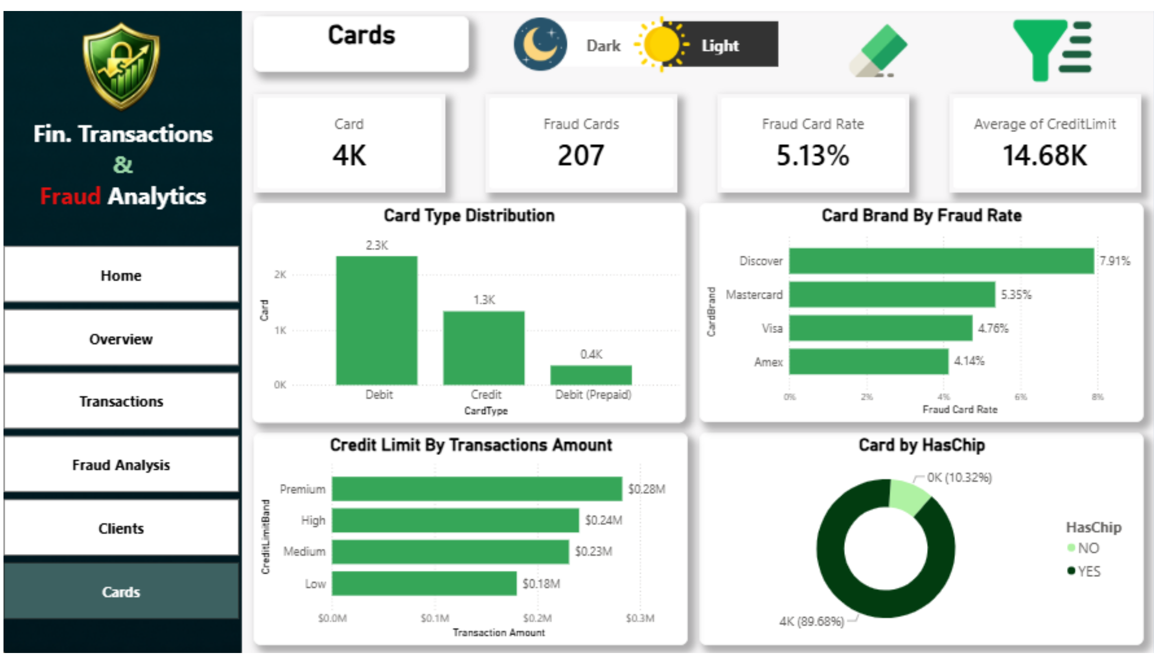
4- Fraud Analysis



5 - Clients



6- Cards



Thank You

Presented By:
Mohamed Saber